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Problem Set #3

Due Date: 02/24/2026

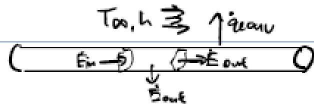
Acknowledgement:

No help

1.

1. Known: $R, \dot{q}, T_\infty, h$

Find: a) Energy balance equation b) governing diff eq. c) BC, IC



Assumptions: 1) No thermal radiation 2) constant k, ρ, c_p

3) constant T_∞ & h 4) r is very small

5) temp. at rod in radial direction uniform

a) $$\dot{E}_{in} + \dot{E}_g - \dot{E}_{out} - \dot{E}_{conv} = \dot{E}_{st}$$

$$\dot{E}_{in} = q_x = -kA\frac{\partial T}{\partial x}\Big|_x$$

$$\dot{E}_{out} = q_{x+dx} = -kA\frac{\partial T}{\partial x}\Big|_x$$

$$\dot{E}_g = \dot{q}(A\,dx)$$

$$\dot{E}_{conv} = h(P\,dx)(T(x,t) - T_\infty)$$

$$\dot{E}_{st} = \rho c_p (A\,dx)\frac{\partial T}{\partial x}$$

$$q_x + \dot{q}A\,dx - q_{x+dx} - hP\,dx(T - T_\infty) = \rho c_p A\,dx\frac{\partial T}{\partial t}$$

b) $$q_{x+dx} = q_x + \frac{\partial q_x}{\partial x}dx$$

$$-\frac{\partial q_x}{\partial x}dx + \dot{q}A - hP(T - T_\infty) = \rho c_p A\frac{\partial T}{\partial t}$$

$$\frac{\partial q_x}{\partial x} = -kA\frac{\partial^2 T}{\partial x^2}$$

$$kA\frac{\partial^2 T}{\partial x^2} + \dot{q}A - hP(T - T_\infty) = \rho c_p A\frac{\partial T}{\partial t}$$

$$k\frac{\partial^2 T}{\partial x^2} + \dot{q} - \frac{2h}{R}(T - T_\infty) = \rho c_p\frac{\partial T}{\partial t}$$

$$\frac{\partial T}{\partial t} = \alpha\frac{\partial^2 T}{\partial x^2} + \frac{\dot{q}}{\rho c_p} - \frac{2h}{\rho c_p R}(T - T_\infty)$$

c) BC $x = 0$

$$T(0,t) = T_1$$

$x = L$

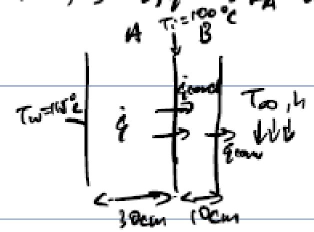
$$-k\frac{\partial T}{\partial x}\Big|_{x=L} = h(T(L,t) - T_\infty)$$

IC $T(x,0) = T_i$ for $0 \le x \le L$

2.

2. Known: $L_A, L_B, k_B, T_\infty, T_w, T_i$

Find: a) T_s b) $\dot{q}$ c) k_A d) Plot graph



Assumptions: 1) Left side of wall A insulated

2) Constant T_∞, h 3) steady state, 1D

a) $q'' = k_B \frac{T_i - T_s}{L_B}$

$q'' = h(T_s - T_\infty)$

$T_s = T_i - \frac{q'' L_B}{k_B}$

$q'' = h\left(T_i - \frac{q'' L_B}{k_B} - T_\infty\right)$

$q''\left(1 + \frac{h L_B}{k_B}\right) = h(T_i - T_\infty)$

$$q'' = \frac{500(100-20)}{\frac{8}{3}} = 15000\ W/m^2$$

$$T_s = T_\infty + \frac{q''}{h} = 20 + \frac{15000}{500} = 50°C$$

b) $q'' = \dot{q} L_A$

$$\dot{q} = \frac{q''}{L_A} = \frac{15000}{0.3} = 50000\ W/m^3$$

c) $$\frac{d^2T}{dx^2} = -\frac{\dot{q}}{k_H}$$

$x=0, \frac{dT}{dx}(0)=0$

$$T_H(x) = T_W - \frac{\dot{q}}{2k_H}x^2$$

$x = L_H, T_H(L_H) = T_i$

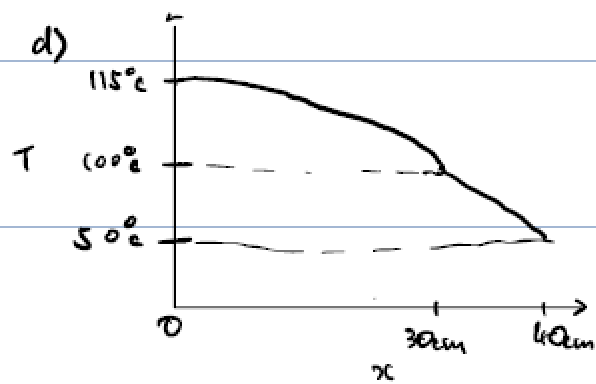
$$T_i = T_W - \frac{\dot{q}}{2k_H}L_H^2$$

$$T_W - T_i = \frac{\dot{q}L_H^2}{2k_H}$$

$$k_H = \frac{\dot{q}L_H^2}{2(T_W - T_i)} = \frac{(50000)(0.3^2)}{2(15)}$$

$$= 150 W/(m\,K)$$

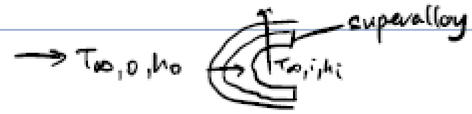
d)



3.

3. Known: $T_{\infty,o}, h_o, T_{\infty,i}, h_i, k_b, L_b, k_{TBC}, L_{TBC}, R''$

Find: a) T_{outer} b) T_{outer}



Assumptions: 1) Radiation neglected 2) steady 1D

3) Constant $T_{\infty,o}, h_o, T_{\infty,i}, h_i$

a) $T_{\infty,o}$ — $\frac{1}{h_o}$ — $T_{s,o}$ — $\frac{L_b}{k_b}$ — $\frac{1}{h_i}$ — $T_{\infty,i}$

$$R''_{tot} = \frac{1}{h_o} + \frac{L_b}{k_b} + \frac{1}{h_i} = \frac{1}{1000} + \frac{0.005}{25} + \frac{1}{500} = 0.001 + 0.0002 + 0.002$$
$$= 0.0032$$

$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{tot}} = \frac{1616-400}{0.0032} = 380\,000\ \text{W/m}^2$$

$$T_{s,o} = T_{\infty,o} - \frac{q''}{h_o} = 1616 - \frac{380000}{1000} = 1236\text{K}$$

1236 > 1200, not maintained below T_{max}

b) $T_{\infty,o}$ — $\frac{1}{h_o}$ — $T^{TBC}_{s,o}$ — $\frac{L_{TBC}}{k_{TBC}}$ — T^{TBC}_{int} — R'' — $T^{b}_{s,o}$ — $\frac{L_b}{k_b}$ — $T^{b}_{s,i}$ — $\frac{1}{h_i}$ — $T_{\infty,i}$

$$R''_{tot} = \frac{1}{h_o} + \frac{L_{TBC}}{k_{TBC}} + R'' + \frac{L_b}{k_b} + \frac{1}{h_i}$$

$$= 0.001 + 0.0005 + 0.0001 + 0.0002 + 0.002$$

$$= 0.0038$$

$$q'' = \frac{1616-400}{0.0038} = 320\,000\ \text{W/m}^2$$

$$T^{TBC}_{s,o} = 1616 - \frac{q''}{h_o} = 1616 - \frac{320000}{1000} = 1296\text{k}$$

$$\Delta T_{TBC} = q''\frac{L_{TBC}}{k_{TBC}} = 320000(0.0005) = 160$$

$$T^{TBC}_{int} = 1296 - 160 = 1136\text{k}$$

$$\Delta T_{contact} = q''R'' = 320000(10^{-4})$$
$$= 32$$

$T^{b}_{s,o} = 1136 - 32 = 1104\text{k}$, maintained below T_{max}